
Holistically Evaluating Legal Reasoning in Large Language Models

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Abstract

The increasing capacities and prominence of Large Language Models (LLMs) lends itself to applications across various industries. In particular, increased adoption of LLMs to the legal field showcases the potential to increase accessibility to legal resources, alleviate lawyer burden, and reason through sophisticated tasks.

The existing benchmark for evaluating an LLM’s legal reasoning capacity, *LegalBench*, has been criticized for various issues. *LegalBench* tasks, which center around civil law, do not reflect the heterogeneous language use of the law; they also do not deal with legal ambiguity. The result is that there is no realistic standard of evaluation for longer, more complex tasks.

Notably, the *LegalBench* benchmark framework independently tests LLM capabilities across tasks in five categories: issue-spotting, rule recall, rule application & conclusion, interpretation, and rhetorical understanding. This misrepresents a model’s true capabilities, as to produce a correct answer in a real-world scenario, the model must correctly work across all five subdivisions simultaneously. This project will explore the ways in which we can more accurately evaluate LLM capabilities in the domain of legal reasoning through a more comprehensive testing approach.

1 Goals and Objectives

Our ultimate goal is to determine whether we can test the most advanced language models against a more robust legal reasoning benchmark. Models tested will include commercial API-access models (including GPT-4, GPT-3.5, and Claude-1), and open source models of different size points (those of 3B parameters, 7B parameters, and 13B Parameters). Tentative open source-models for study include the following:

- Together: Incite-Instruct-7B, Incite-Base-7B, Incite-Instruct-3B
- Meta OPT: OPT-2.7B, OPT-6.7B, OPT-13B
- TII Falcon: Falcon-7B-Instruct
- MosaicML MPT: MPT-7B-8k-Instruct
- LMSYS Vicuna: Vicuna-7B-16k, Vicuna-13B-16k
- Google FLAN-T5: Flan-T5-XL, Flan-T5-XXL
- Meta LLama-2: LLaMA-2-7B, LLaMA-2-13B

- Wizard: WizardLM-13B
- BigScience BLOOM: BLOOM-3b and BLOOM-7B

The first objective is to create some subset of the open-source *Statutory Reasoning Assessment* dataset (SARA) to evaluate the ability of models to apply legal reasoning to real-world scenarios. Tasks should be structured to require the model to engage in all five reasoning categories within a single scenario.

The second phase of the project will involve prompt experimentation - the goal being to demonstrate how we can induce more coherent legal reasoning. This will include performance records with zero-, one-, and few- shot prompting (and, time permitting, chain-of-thought prompting).

Performance review will require design of a streamlined evaluation criteria that can synthesize model outputs in bulk. In addition to a performance review, this thesis will produce suggested modeling approaches for future improvement.

2 Deliverables and Timeline

For this project, I will work with a minimum of one open-source dataset; clean and modify this set accordingly, and experiment with how we can induce more comprehensive legal reasoning in LLMs. I will publish the modified dataset, scripts, and formulas for evaluation as a GitHub repository; I will also create a single-access point for this information via a webpage.

The tentative timeline for this thesis is as follows, with the italicized dates serving as the task date of completion:

I. Task Design & Modification

February 13: Obtain access to the various LLMs to be tested; research legal evaluation criteria; begin work on thesis infrastructure (i.e. GitHub repository, webpage, etc.)

February 27: Access and modify SARA scripts and dataset; formulate robust experiment plan; finalize model list.

II. Structured Questioning & Scoring Mechanism Design

March 6: Experimentation and documentation of zero-shot prompting

April 3: Generation of one- and few-shot prompts, experimentation with prompt choice, and documentation.

III. Model Evaluation & Refinement, Report Production

April 10: Evaluate comprehensiveness of testing. Time permitting, add chain-of-thought prompting and/or fine-tuning component evaluation.

April 25: Completion of online documentation, final report, and poster.

References

[1] *A Dataset for Statutory Reasoning in Tax Law Entailment and Question Answering* [paper] [bibtex] Nils Holzenberger & Andrew Blair-Stanek & Benjamin Van Durme. Proceedings of the 2020 Natural Language Processing (NLLP) Workshop, 24 August 2020, San Diego, United States.

[2] *LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models*. Neel Guha et al. *arXiv [Cs.CL]*, 2023, <http://arxiv.org/abs/2308.11462>. arXiv.

[3] *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*. Wei, Jason, et al. *arXiv [Cs.CL]*, 2023, <http://arxiv.org/abs/2201.11903>. arXiv.